Maximum Subarray by Divide and Conquer

Topic: DivideAndConquer File: d4.pdf

PROBLEM

Find the contiguous subarray with the greatest sum.

ALGORITHM

Recursively find the best subarray in the left half and right half. Also compute the best crossing sum from the middle, then take the maximum of the three.

PSEUDOCODE

```
best(A, lo, hi):
  if lo == hi: return A[lo]
  mid = (lo + hi) // 2
  return max(best(left), best(right), best_crossing(A, lo, mid, hi))
```

EXAMPLE

For [-2,1,-3,4,-1,2,1,-5,4], the maximum-sum subarray is [4,-1,2,1] with sum 6.

COMPLEXITY

The recurrence is T(n) = 2T(n/2) + O(n), giving O(n log n) time and O(log n) recursion depth.

NOTE

Kadane’s algorithm solves the same problem in O(n) time, but this exercise demonstrates divide and conquer.